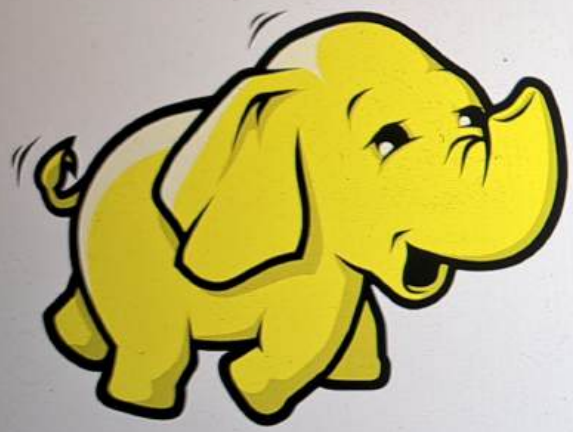

In this lecture

- Data processing software applications
- Evolution of Data
- Big Data Problems and Solutions
- Hadoop Platform

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Software Engineering Specializations

1. System Software
3. Programming Tools
3. Desktop Applications
4. Data Processing Application
5. Web & Mobile Applications
6. Software Platforms
7. Cloud Computing
8. Machine Learning and AI

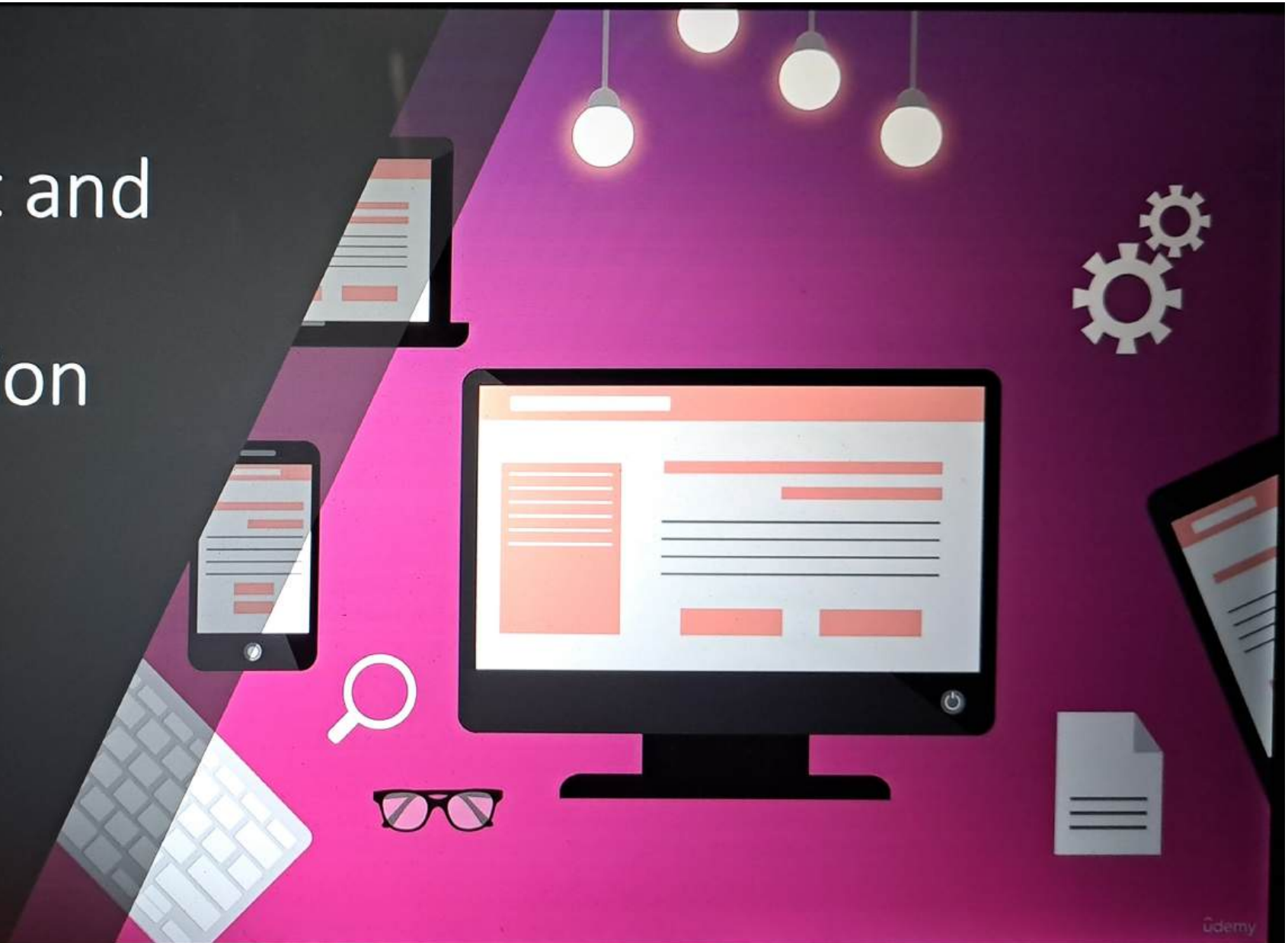
Beginning of Business Data Processing

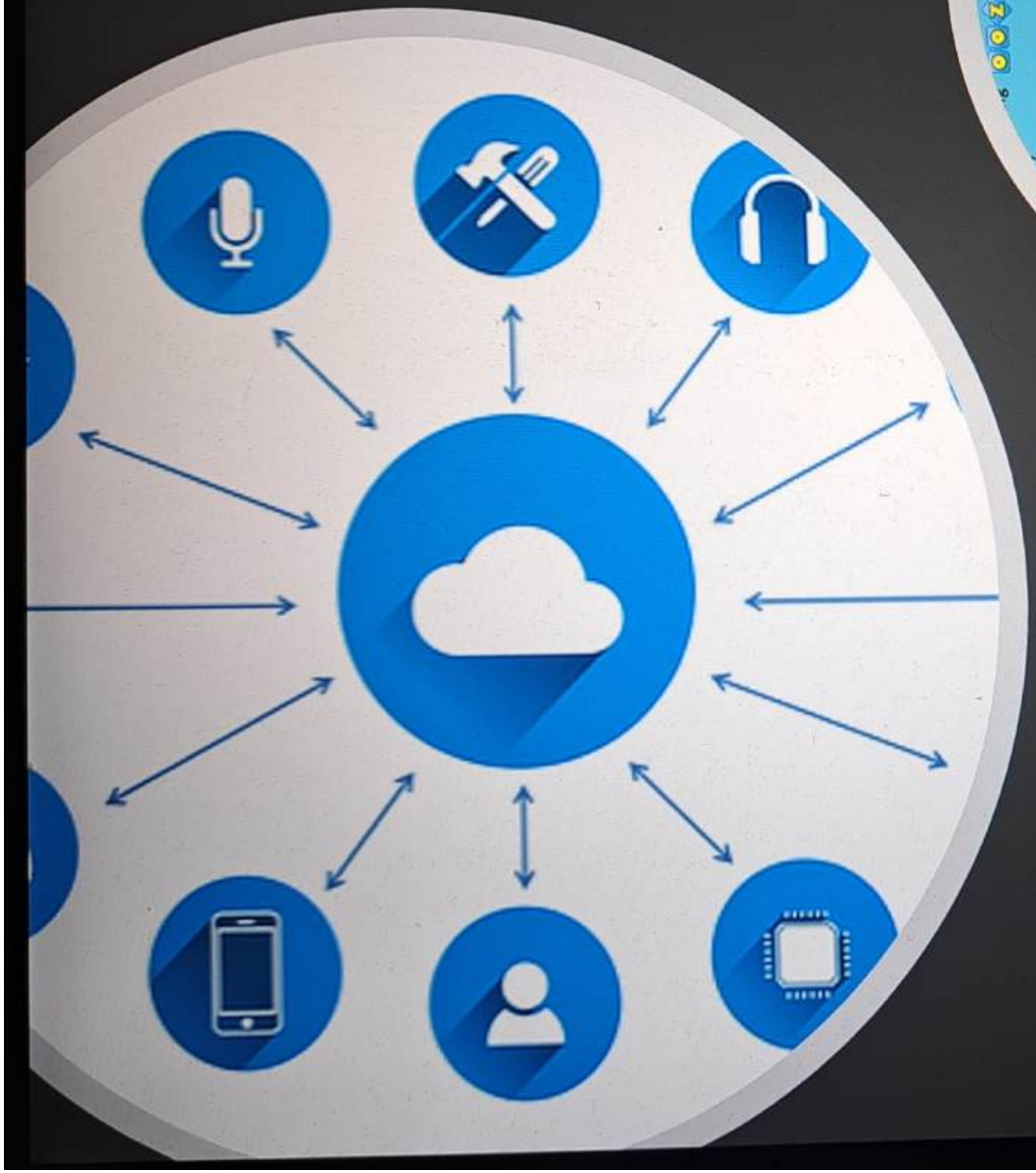
COBOL

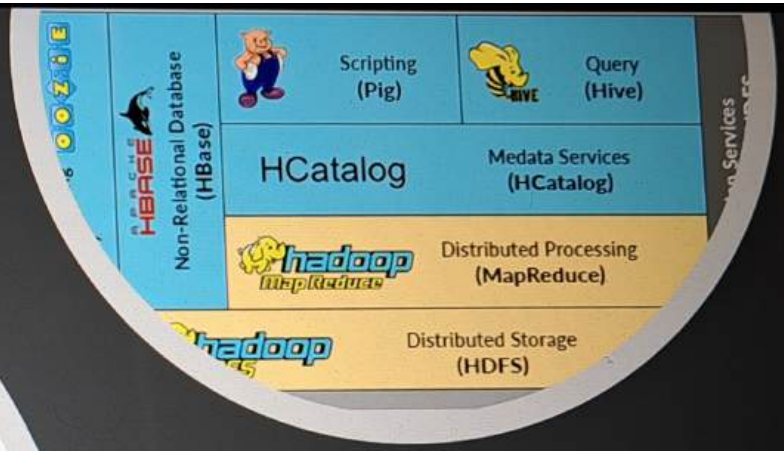
- Common Business-Oriented Language
- Used in business and finance for companies and governments
- Allowed to store data in files, create index files, and process data efficiently
- Widely used on mainframe computers
- Used for large-scale batch and transaction processing jobs

Revolution in Business Data Processing

Internet and Mobile Revolution







Platform Revolution

RDBMS Systems

Exadata Appliance

Teradata Appliance

COBOL

Programing Language

Business Data Processing Beginning and Evolution

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RDBMS - What do they offer?

- SQL - An easy Data Query Language
- Scripting Languages such as PL/SQL and Transact SQL
- Interface for other programming languages such as JDBC and ODBC

person_ID	name	first	last	middle	email	phone	fax	title
3130	Burks, Rosella	Rosella	Burks		BurksR@univ.edu	963.555.1253	963.777.4065	Professor
3297	Avila, Damien	Damien	Avila		AvilaD@univ.edu	963.555.1352	963.777.7914	Professor
3547	Olsen, Robin	Robin	Olsen		OlsenR@univ.edu	963.555.1378	963.777.9262	Assistant Professor
1538	Moises, Edgar Estes	Edgar	Moises	Estes	MoisesE@univ.edu	963.555.2731x3565	963.777.8264	Professor
2941	Brian, Heath Pruitt	Heath	Brian	Pruitt	BrianH@univ.edu	963.555.2800	963.777.7249	Associate Curat
2401	Claude, Elvin Haney	Elvin	Claude	Haney	ClaudeE@univ.edu	963.555.2902	963.777.9730	Curator
2070	Mosley, Edmund	Edmund	Mosley		MosleyE@univ.edu	963.555.2945	963.777.9285	Assistant Pro
2561	Derek, Antoine Mccoy	Antoine	Derek	Mccoy	DerekA@univ.edu	963.555.2992	963.777.5454	Curator
1625	Hawkins, Callie	Callie	Hawkins		HawkinsC@univ.edu	963.555.3350x6480	963.777.4949	Professor
1307	Pate, Andrea	Andrea	Pate		PateA@univ.edu	963.555.3723	963.777.5049	Professo

Table – Row/Column

Structured Data

```
person_ID,name,first,last,middle,email,phone,fax,title
3130,"Burks, Rosella ",Rosella,Burks,,BurksR@univ.edu,963.555.1253,963.777.4065,Professor
3297,"Avila, Damien ",Damien,Avila,,AvilaD@univ.edu,963.555.1352,963.777.7914,Professor
3547,"Olsen, Robin ",Robin,Olsen,,OlsenR@univ.edu,963.555.1378,963.777.9262,Assistant Profes
1538,"Moises, Edgar Estes",Edgar,Moises,Estes,MoisesE@univ.edu,963.555.2731x3565,963.777.82
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1625,"Hawkins, Callie ",Callie,Hawkins,,HawkinsC@univ.edu,963.555.3350x6480,963.777.49
1307,"Pate, Andrea ",Andrea,Pate,,PateA@univ.edu,963.555.3723,963.777.5049,Professor
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File – Row/Column

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Semi-structured Data

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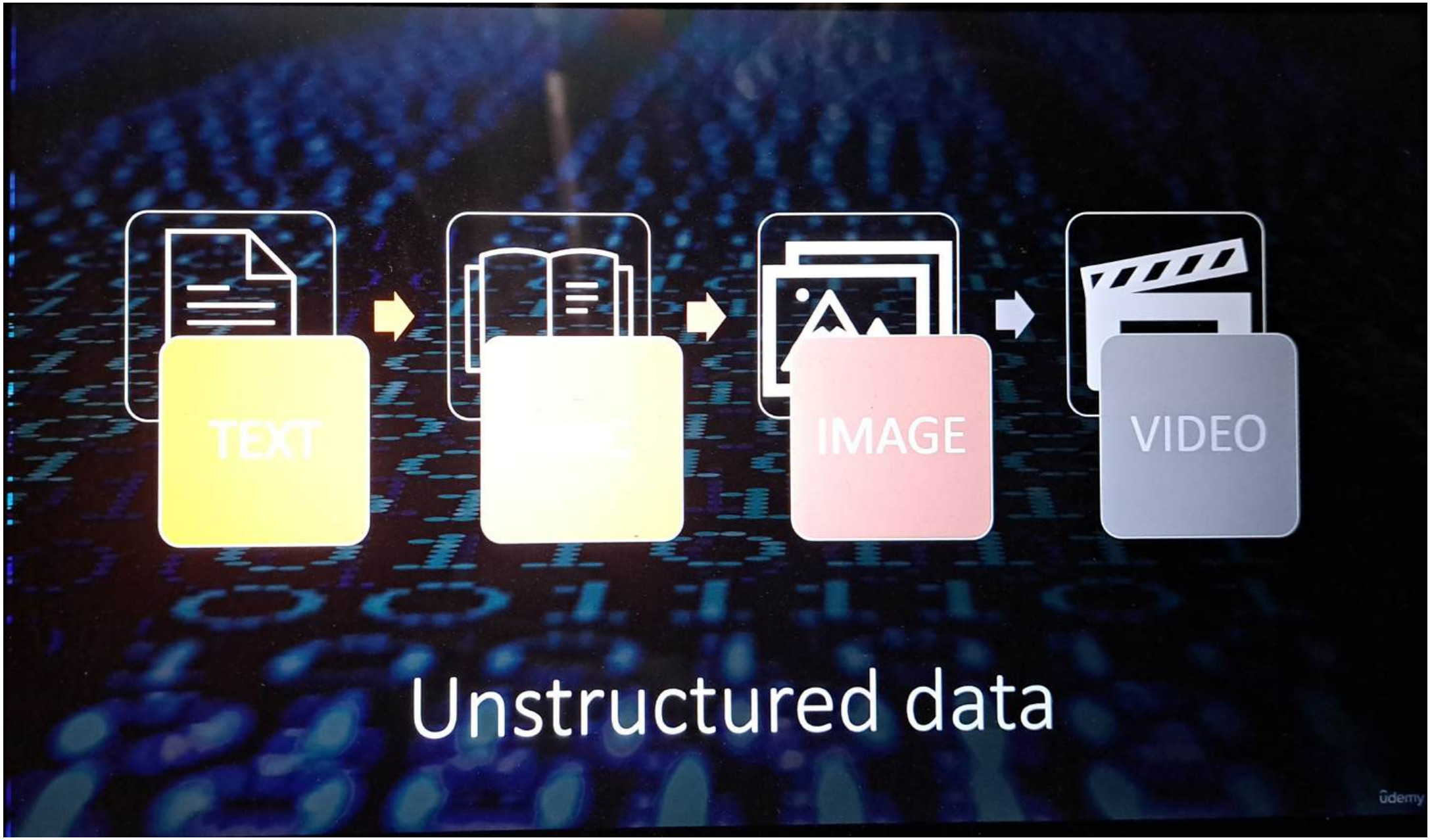
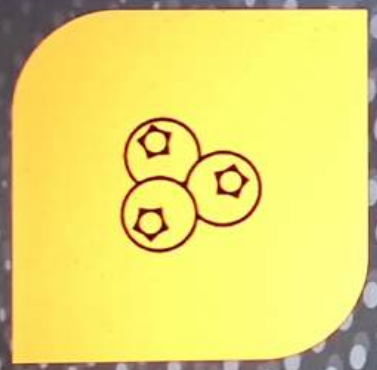

TEXT
IMAGE
VIDEO
Unstructured data
Udemy

Data Categories

1. Structured
2. Semi-Structured
3. Un-Structured

Big Data Problem



VARIETY

VOLUME

VELOCITY

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Big Data

Platform Requirements

Store High Volumes of data arriving at a higher velocity

Accommodate structured, semi-structured, and unstructured data variety

Process high volumes of a variety of data at a higher velocity

Summary

- Data processing is one of the critical business requirements
- We used RDBMS for decades to develop data processing applications
- The advent of the internet started putting following data processing challenges
  - We started collecting a Variety of Data
    - Structured data
    - Semi-Structured data
    - Un-Structured data
  - Businesses started collecting high Data Volumes
  - Need to collect and process data at a high velocity
- The new data challenge is popularly known as Big Data Problem
- The Big Data problem was defined using the 3Vs of Big data - Variety, Volume, and Velocity
- RDBMS failed to handle the Big Data problem
- Industry needed a new approach or platform to handle the Big Data Problem

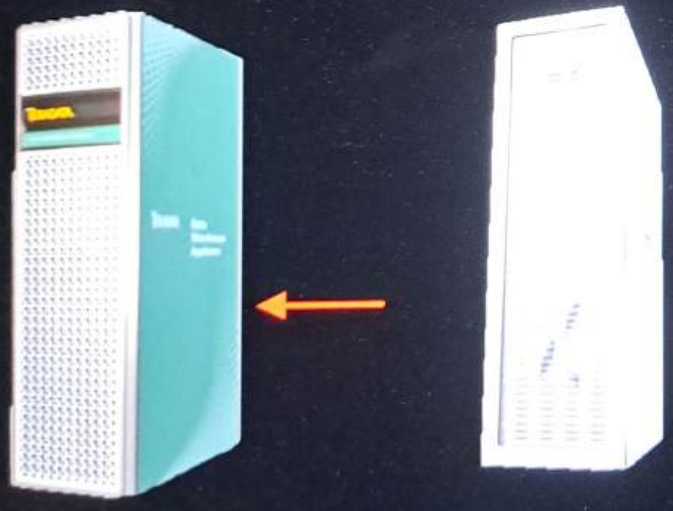
Two Approaches of
Big Data Solution

- Monolithic Approach
- Distributed Approach

Approaches of Big Data Solution

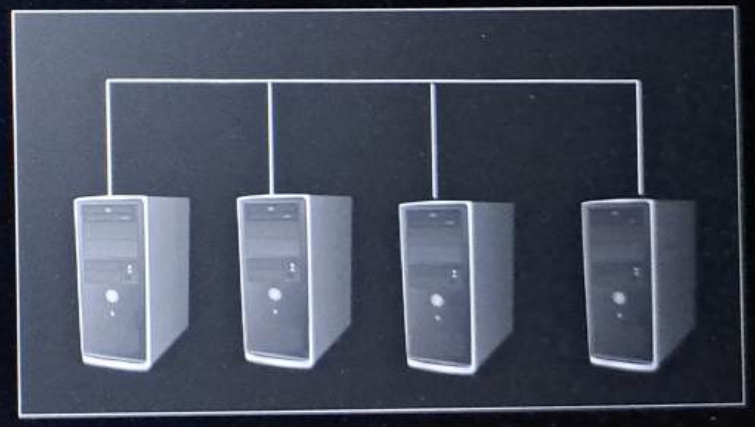
Monolithic Approach

Massive Resource: CPU, RAM, Disk



Distributed Approach

Cluster
Resource Pool: CPU, RAM, Disk



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Approaches of Big Data Solution

Criteria

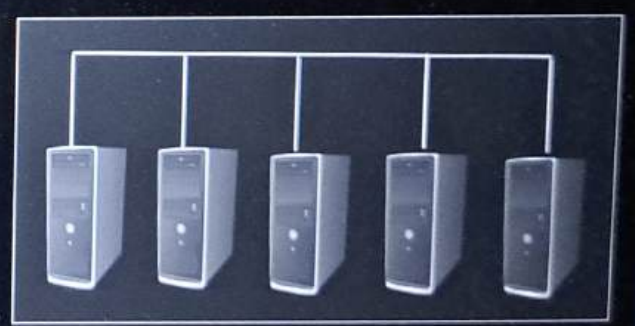
1. Scalability
2. Fault Tolerance and HA
3. Cost-Effectiveness

Monolithic Approach

1. Vertical
2. Primary/Secondary
3. Expensive

Distributed Approach

1. Horizontal
2. Multifold
3. Economical



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- RDBMS failed to handle the Big Data problem
- Industry needed a new approach or platform to handle the Big Data Problem
- Hadoop came as a distributed Big Data processing platform

Hadoop Platform